PRINCETON UNIVERSITY PRESS

Chapter Title: B. PROBABILITY DENSITY FUNCTIONS

Book Title: Natural Complexity

Book Subtitle: A Modeling Handbook

Book Author(s): Paul Charbonneau

Published by: Princeton University Press

Stable URL: https://www.jstor.org/stable/j.ctt1vwmgzt.17

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Princeton University Press is collaborating with JSTOR to digitize, preserve and extend access to *Natural Complexity*

JSTOR

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

B

PROBABILITY DENSITY FUNCTIONS

Probability density functions (PDFs) measure the probability of finding a measurement in some specified interval of possible values for the measured quantities. As analysis and interpretative tools, they are used repeatedly in this book, and a basic understanding of their construction and interpretation is essential. This is the aim of this appendix. Appendix B.1 introduces the idea at the pre-calculus level through a simple example, while the following sections require a working knowledge of the calculation of derivatives and integrals of functions of a single variable.

B.1 A Simple Example

The following numbers are the grades (in percent) obtained a few years ago by my cohort of $N = 83$ undergraduate students on the midterm exam of my Introduction to Computational Physics class:

[46, 84, 70, 66, 41, 82, 69, 59, 28, 81, 88, 82, 83, 33, 27, 51, 62, 72, 87, 55, 66, 68, 55, 86, 75, 74, 56, 81, 60, 44, 84, 86, 75, 34, 96, 45, 57, 79, 81, 52, 24, 38, 74, 89, 68, 85, 85, 45, 62, 96, 45, 40, 48, 90, 46, 57, 33, 71, 67, 82, 94, 43, 16, 88, 46, 91, 82, 55, 71, 86, 77, 63, 81, 78, 59, 84, 100, 69, 92, 69, 44, 64, 88].

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

Let g_k represent the grade obtained by the kth student. The class average, $\langle g \rangle$, for this exam is simply given by the sum of all grades divided by the class size:

$$\langle g \rangle = \frac{1}{N} \sum_{k=1}^{N} g_k, \tag{B.1}$$

which for the above data is 66.3. To what degree is this number really representative of students' grades? This information can be obtained by constructing the PDF of the grades.

For such a discrete data set, an approximation to the PDF can be built by constructing a *histogram*. This consists in dividing the allowed range of the measured variable—here grades between 0% and 100%—into contiguous bins, each spanning a range of grades, and then counting how many data points fall into each bin, for example counting how many students have a grade between 60% and 64.99%, or between 65% and 69.99%. This defines a discrete function

$$h_m(g; b), \quad m = 1, \ldots, M, \tag{B.2}$$

where b is the bin size and h_m is the count in the mth bin. The number of bins is simply $M = 100/b$, i.e., the numerical extent of the data, here 100, divided by the bin size.

Figure B.1 shows histograms of the above data, for bin sizes of 2, 5, and 10, and with the class average indicated by the vertical dashed-line segment.[1] The heights of the histogram bins obviously depend on the choice of bin size; for a data set of a given length, the wider the bins the higher the corresponding counts, and some bins can of course remain empty. No matter the bin size, in all cases the sum of counts in all the bins is always equal to the class size $N = 83$; in other words, the histogram can be

[1]Since a histogram is, fundamentally, a discrete function of the measurement variable, it is customary to plot it in so-called histogram mode, i.e., as a piecewise-constant function, varying discontinuously at bin boundaries.

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

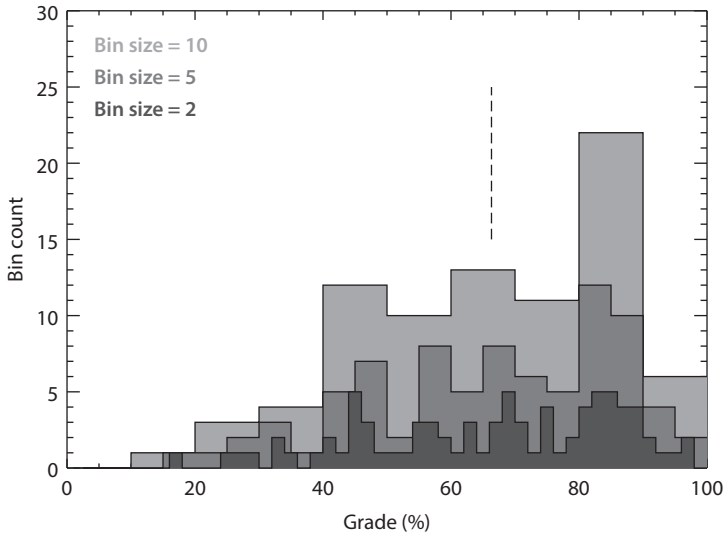

Figure B.1. Histograms for my midterm exam data set, for bin sizes of 2 (dark gray), 5 (gray), and 10 (light gray). The vertical dashed-line segment is the class average computed using equation (B.1).

normalized:

$$\frac{1}{N}\sum_{m=1}^{M} h_m = 1. \tag{B.3}$$

Knowing h_m also allows an alternative procedure to compute the class average:

$$\langle g \rangle = \frac{1}{N}\sum_{m=1}^{M} h_m \times b_m, \tag{B.4}$$

where b_m is the grade value at the center of bin m.

Python's NumPy library includes a function, named `numpy.histogram()`, that accepts as input an array of values, such as my midterm exam grades above, and returns an array containing histogram bin counts (10 equidistant bins by

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

default). It is also possible to set the number of bins, their sizes and ranges through the function's argument list; see the SciPy/NumPy documentation (URLs provided at the end of appendix A).

To turn the counts of figure B.1 into a probability, we need to divide them by the class size $N = 83$, and to turn them into a probability *density* we also need to divide by the bin size. This last step is required so that the quantity $h_m \times b$ measures the probability p of finding, in the grade data set, a grade falling between the bounding values of each corresponding bin. This is the very definition of a probability density. Figure B.2 shows the result of this procedure for the histogram of bin size $b = 5$ from figure B.1, and defines the *discrete* PDF for this data set. Its detailed shape is obviously influenced by the chosen bin size, and some of the finer structure also reflects specificities of the underlying data set; figure B.2 would not be identical if I had used midterm exam data from a different year, even though my average grade for the midterm exam in this course always hovered around 65%. The finite size of the data set also guarantees that the PDF will not be smooth, and this would remain the case even if I opted to minimize my grading time by assigning purely random grades in the allowed range (more on this point in appendix C.2 below). But since I did not engage in such accelerated grading practices, the PDF of figure B.2 does capture something about students' performance (as well as my grading performance, presumably). Note how the most probable grade, i.e., the bin with the highest probability density (0.0289 for the bin 80–85%), is not the one spanning the average grade (bin 65–70%, with a probability density of 0.0193), reflecting the fact that this distribution is asymmetric about its mean value. I observed such an asymmetry almost every year I taught this course; it was something real.[2]

[2]In fact, my grade PDFs often could be reasonably well fit by a combination of two Gaussians, one very broad and centered around 60%, the other much

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

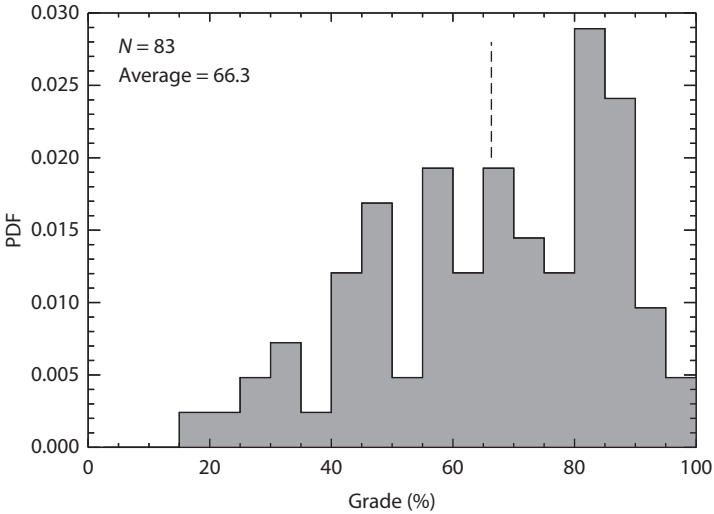

Figure B.2. Discrete PDF constructed from the gray histogram of figure B.1 (bin size = 5). The shapes are, of course, identical on both plots; only the vertical scale is altered, from raw counts to probability density.

B.2 Continuous PDFs

In the limit of very large data sets and infinitesimally small bin size (in the calculus sense), the PDF can be considered a smooth, continuous function $f(x)$, where the quantity $f(x)\,\mathrm{d}x$ measures the probability of finding a measurement in the interval $[x, x + \mathrm{d}x]$. The equivalent of (B.3) becomes the normalization

narrower and centered around 80%. These could be traced pretty directly to two distinct groups of students in the class: physics major students, taking the course in their first semester, and physics-math double-major students, taking the class in the third semester. The latter group dominated the PDF peak around 85%, indicating rather unambiguously that my midterm exams favored them unfairly. Seeing this pattern repeat itself year after year, I eliminated the midterm exam from the course evaluation.

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

constraint

$$\int f(x)\,\mathrm{d}x = 1, \tag{B.5}$$

and the equivalent of (B.4) is then

$$\langle x \rangle = \int f(x)x\,\mathrm{d}x. \tag{B.6}$$

In both cases the integral must cover the full range of the variable x.

B.3 Some Mathematical Properties of Power-Law PDFs

Probability density functions of event sizes taking the form of power laws are common in many of the natural and simulated systems considered in this book, so they deserve a bit more attention. Such a power-law PDF is written as

$$f(x) = f_0 x^{-\alpha}, \quad \alpha > 0, \quad x \in [x_0, x_M], \tag{B.7}$$

where the constant f_0 is set by the normalization constraint (B.5). Without loss of generality we can set the normalization interval of the PDF to the range $[1, \infty]$, so that the substitution of (B.7) into (B.5) yields

$$f_0 = \lim_{x_M \to \infty} \frac{\alpha - 1}{1 - (x_M)^{1-\alpha}}. \tag{B.8}$$

Evidently, the normalization is only possible provided $\alpha > 1$, otherwise $f_0 \to \infty$ in the limit $x_M \to \infty$. For normalizable PDFs we have

$$f_0 = \alpha - 1, \quad \alpha > 1, \tag{B.9}$$

in the $x_M \to \infty$ limit. Now consider a situation where the variable x is extracted from measurements spanning the range $[x_0, x_M]$, as described by a normalizable power-law PDF ($\alpha > 1$). In such a situation the normalization constant becomes $f_0 = (\alpha - 1)/(x_0^{1-\alpha} - x_M^{1-\alpha})$, and the mean value of the variable

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

calculated via equation (B.6) is

$$\langle x \rangle = \frac{f_0}{\alpha - 2}(x_0^{2-\alpha} - x_M^{2-\alpha}), \quad (\alpha \neq 2). \tag{B.10}$$

In many cases the PDF spans many orders of magnitude in the variable x, i.e., $x_0 \ll x_M$. We can then distinguish two regimes:

- $1 < \alpha < 2$. This implies $2 - \alpha > 0$, so that the term involving x_M dominates in (B.10). We then have

$$\langle x \rangle = \frac{f_0}{2 - \alpha} x_M^{2-\alpha}. \tag{B.11}$$

- $\alpha > 2$. This implies $2 - \alpha < 0$, so that the term involving x_0 dominates (B.10), in which case,

$$\langle x \rangle = \frac{f_0}{\alpha - 2} x_0^{2-\alpha}. \tag{B.12}$$

The special case $\alpha = 2$ is "left as an exercise," as we like to say in the business.

These two distinct regimes, as delineated by the value of the power-law index α, have important consequences when constructing a PDF from a finite set of individual measurements. Note in particular that, for $\alpha < 2$, the average event size is determined by the largest measured event. These being rare if the PDF is a power law, computing the mean event size from an experimental or numerical data set containing too few events could lead to a gross underestimate of the mean value. This is no longer the case if $\alpha > 2$, since the mean value is then dominated by the smaller, more frequent events, which will be well represented even in a (relatively) small data set. If $\alpha < 1$, the mean value cannot even be mathematically defined.

B.4 Cumulative PDFs

Sometimes, observational data are represented through a *cumulative* PDF $f(> x)$, such that $f(> x)\,\mathrm{d}x$ measures the probability

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

of finding a measured value larger than x. Our encounter with the Gutenberg–Richter law in chapter 8 offered one example. If x is distributed as a power law, then we have

$$f(> x) = \int_x^\infty f_0 x^{-\alpha}\,\mathrm{d}x = x^{1-\alpha}, \tag{B.13}$$

where the second equality holds only if the distribution can be normalized, requiring $\alpha > 1$, so that $f_0 = \alpha - 1$. In such a situation, the cumulative PDF is also a power law, with an index differing by unity as compared to the index of the usual noncumulative PDF.

B.5 PDFs with Logarithmic Bin Sizes

If the PDF of a measured variable takes the form of a power law, the tail of a PDF constructed from measurements will contain very few events, and so will be very "noisy," making it difficult to reliably infer the numerical value of the power-law exponent α. This is illustrated in figure B.3, for PDFs constructed from a set of $N = 300$ data points extracted from a power-law distribution with index $\alpha = 1.75$. In panel A, the PDF uses a bin size $b = 10$ and is plotted using the usual linear axes. Because the PDF falls off very rapidly with increasing x, here most points end up concentrated in the first bin ($0 \leq b < 10$). When replotting the same data using logarithmic axes, as in panel B, bins for $x > 100$ either contain only one point, or none at all. Fitting a straight line to this PDF looks like a pretty risky proposition. Turning to the cumulative version of the PDF, as shown in figure B.3C, improves the situation somewhat, in that the middle of this distribution could conceivably be fit with a straight line to yield the exponent $\alpha - 1$ (= 0.75 here, and indicated by the dashed line segment). However, choosing the start and end points of the fitting regions will be very tricky unless the PDF spans many orders of magnitude in the measured variable.

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

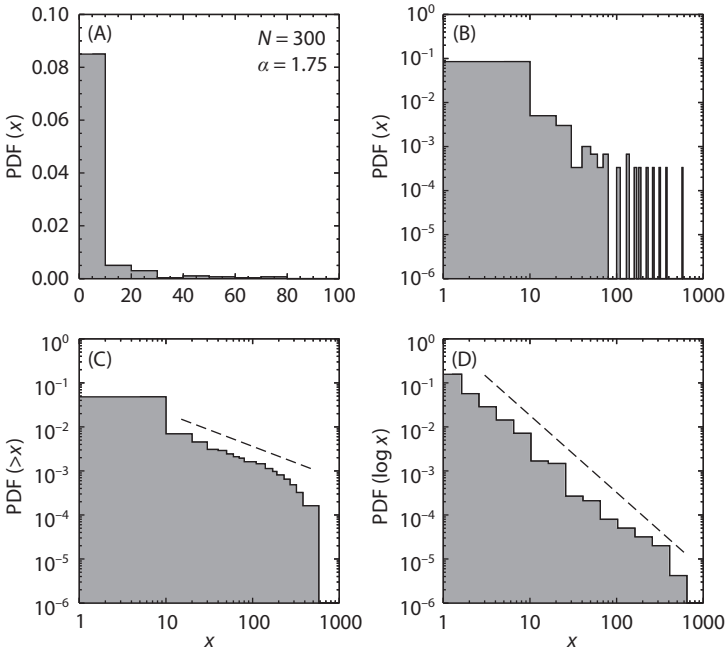

Figure B.3. Plotting can make such a difference. PDFs for the same data are plotted in these four panels, built from a set of 300 synthetic data points extracted from a power-law distribution with index $\alpha = 1.75$. Panel A plots the PDF on a standard plot with linear horizontal and vertical axes, with a bin size of 10 used in constructing the PDF. Panel B plots the exact same data, now using logarithmic axes, and panel C plots the cumulative PDF of the same data, again plotted on logarithmic axes. The resolutely reckless could consider fitting a straight line through the middle decade of the distribution. Panel D plots a PDF of the same data still, now constructed using a logarithmic bin size of $\log(b) = 0.2$ and again plotted on logarithmic axes. A straight-line fit can now be contemplated with some measure of confidence. The dashed lines indicate the true slope of the distribution from which the $N = 300$ data were drawn (minus unity for the cumulative PDF in panel C, as per equation (B.13)).

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

One way around this difficulty is to introduce bin sizes that increase with x. A particularly simple way to achieve this is to construct the histogram function of the *logarithm* of the measurement variable:

$$y = \log(x) \rightarrow \mathrm{d}y = \frac{\mathrm{d}x}{x}, \tag{B.14}$$

and then use a bin size that is constant in y. Whatever the variable we use to construct the PDF, the normalization constraints (B.5) always hold, so that

$$\int f(x)\,\mathrm{d}x = \int f(y)\,\mathrm{d}y = \int \frac{f(y)}{x}\,\mathrm{d}x, \tag{B.15}$$

using the second equality resulting from equation (B.14). We thus conclude that

$$f(x) = \frac{f(\log x)}{x}. \tag{B.16}$$

In other words, we are correcting the counts (and associated probabilities) by accounting for the fact that the bin size increases linearly with x. In this way, even if the PDF is constructed as $f(\log x)$ with logarithmically constant bins, for a power-law PDF, we find that plotting $f(\log x)/x$ versus x using logarithmic axes will still yield a straight line, with slope corresponding to the index α of the underlying PDF $\propto x^{-\alpha}$ for the original measurement variable. Figure B.3D shows the result of this procedure, using the same underlying data as in panels A and B. A linear fit (on the log-log plot representation) can now be envisioned with some measure of optimism.

From figures 4.8 to 8.7, all power-law PDFs encountered in this book are constructed and plotted in this manner. When the measured variable spans many orders of magnitude and N is very large, a straight linear least-squares fit can then return a reasonably accurate estimate of the power-law index α. If either of these conditions is not satisfied however, the inferred value

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

of α may deviate significantly from the underlying "true" value; fortunately, it is (relatively) easy to do better.

B.6 Better Fits to Power-Law PDFs

Compared to the dashed line indicating the true logarithmic slope of the distribution from which the synthetic data were drawn, the PDF in figure B.3 looks pretty good in a "chi-by-eye" sense. Yet a formal linear least-squares fit with equal weight assigned to each bin yields $\alpha = 1.697 \pm 0.051$, somewhat lower than the true underlying value $\alpha = 1.75$. Maybe that's "good enough," or maybe not. Remember the dangers of earthquake prediction encountered at the end of chapter 8.

A proper statistical approach to this fitting problem would be to infer the index for the power-law distribution that has the highest likelihood of having generated the N measured data. If x is a continuous variable and its PDF is normalizable (i.e., $\alpha > 1$), this maximum likelihood estimator for α is

$$\alpha = 1 + N \left(\sum_{n=1}^{N} \ln \frac{x_n}{x_{\min}} \right)^{-1}, \tag{B.17}$$

where $x_{\min}$ is the lower bound of the range within which power-law behavior holds, as determined empirically from the data or on theoretical grounds. The associated standard error (σ) on α is given by

$$\sigma = \frac{\alpha - 1}{\sqrt{N}}. \tag{B.18}$$

For the synthetic data of figure B.3 and with $x_{\min} = 1$, the above expressions yield $\alpha = 1.744 \pm 0.043$, much closer to the target 1.75 than a linear least-squares fit to the log-log plot of logarithmically binned data in panel D.

In working with real data (including measurements from lattice-based simulations), two difficulties must be dealt with. The first is the choice of $x_{\min}$ in equation (B.17). Consider, for

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

example, the PDF of percolation cluster sizes plotted in figure 4.8. As argued in section 4.5, the self-similar fractal structure cannot be expected to extend down to clusters of size 1. Looking at figure 4.8, picking $x_{\min} = 10$ might be a reasonable "chi-by-eye" choice; a higher value would obviously be safer, but this would also mean running the risk of throwing away more potentially useful data, a problem that can become ever more acute the steeper the power law. Here again, statistically sound approaches are available to pick a proper $x_{\min}$ (see the bibliography at the end of this appendix).

A second potential difficulty arises from the fact that the upper end of the distribution of percolation cluster sizes is likely to be affected to some extent by the finite size of the lattice, in the sense that the size of the PDF cannot be expected to drop instantaneously to zero at exactly the largest percolation cluster size $p_c N^2/2$. Different strategies exist to augment such power laws with an upper cutoff, finite-size scaling function. Its defining parameters must then be fit simultaneously with those of the power law, which usually results in a nonlinear fitting problem even if carried out in log-log space. See, for example, the book by Christensen and Moloney cited in the bibliography of chapter 4.

This second difficulty usually does not arise when working with real-world data, for which a hard upper limit is seldom expected. For example, the largest earthquake ever measured, the May 22, 1960 earthquake in Chile, scored 9.5 on the Richter magnitude scale; yet nothing in plate tectonics precludes, in principle, more energetic earthquakes; they simply have not occurred since the beginning of the earthquake-magnitude record. Likewise, the solar flare of figure 12.4 is in all likelihood the largest observed during the space era, but the observations of "superflares," up to 10^4 times more energetic, on stars other than the Sun confirms that nothing close to the upper limit has yet been observed on the Sun—and, as with earthquakes, we can only hope it stays that way.

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms

B.7 Further Reading

Most statistics textbooks discuss PDFs at some level. See, for example,

James, F., *Statistical Methods in Experimental Physics*, 2nd ed., World Scientific (2006);

Roe, B.P., *Probability and Statistics in Experimental Physics*, Springer (1992).

On the inference of power-law behavior in experimental data, see

Clauset, A., Shalizi, C.R., and Newman, M.E.J., "Power-law distributions in empirical data," *SIAM Review*, **51**(4), 661–703.

This content downloaded from
152.46.28.190 on Fri, 08 May 2026 09:46:55 UTC
All use subject to https://about.jstor.org/terms